

VxLEARN Networks

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report: Use the Ipconfig Command

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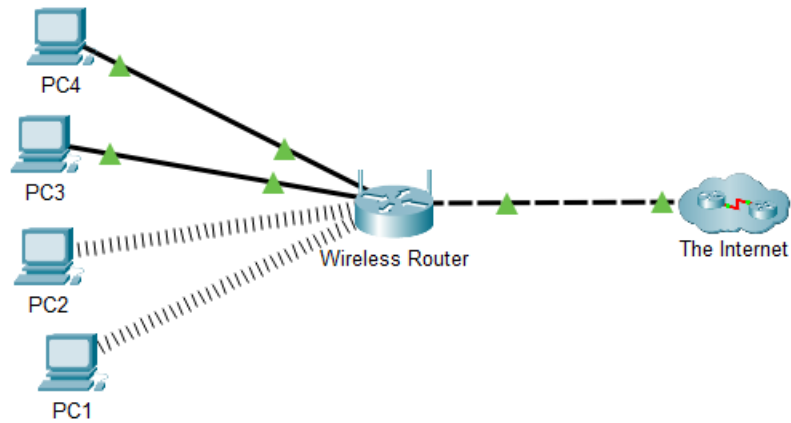
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1. Objectives

- Use the ipconfig command to identify an incorrectly configured PC.

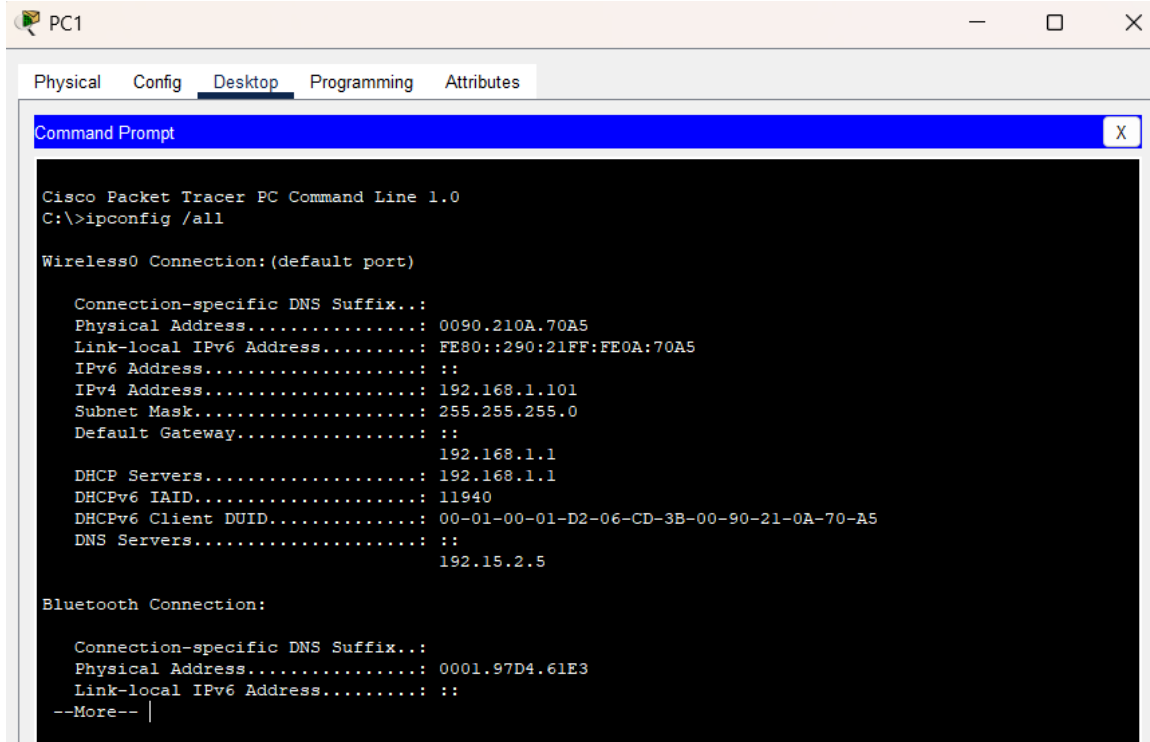
2. Background / Scenario

A small business has four PCs configured with static IP addresses in the 192.168.1.0/24 network. One PC cannot reach the internet or access the website www.cisco.pka. You must determine which PC is misconfigured using the ipconfig /all command.



3. Part 1: Verify Configurations

- Open Command Prompt on each PC.
- Run the command: `ipconfig /all`
- Record the IP address, subnet mask, and default gateway for all PCs.



The screenshot shows a Cisco Packet Tracer window for PC1. The 'Desktop' tab is selected, and the 'Command Prompt' application is open. The command prompt displays the output of the `ipconfig /all` command, showing configuration details for both Wireless0 and Bluetooth connections. The Wireless0 connection is the active network interface, showing an IPv4 address of 192.168.1.101, a subnet mask of 255.255.255.0, and a default gateway of 192.168.1.1. The Bluetooth connection is also shown but is not active.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

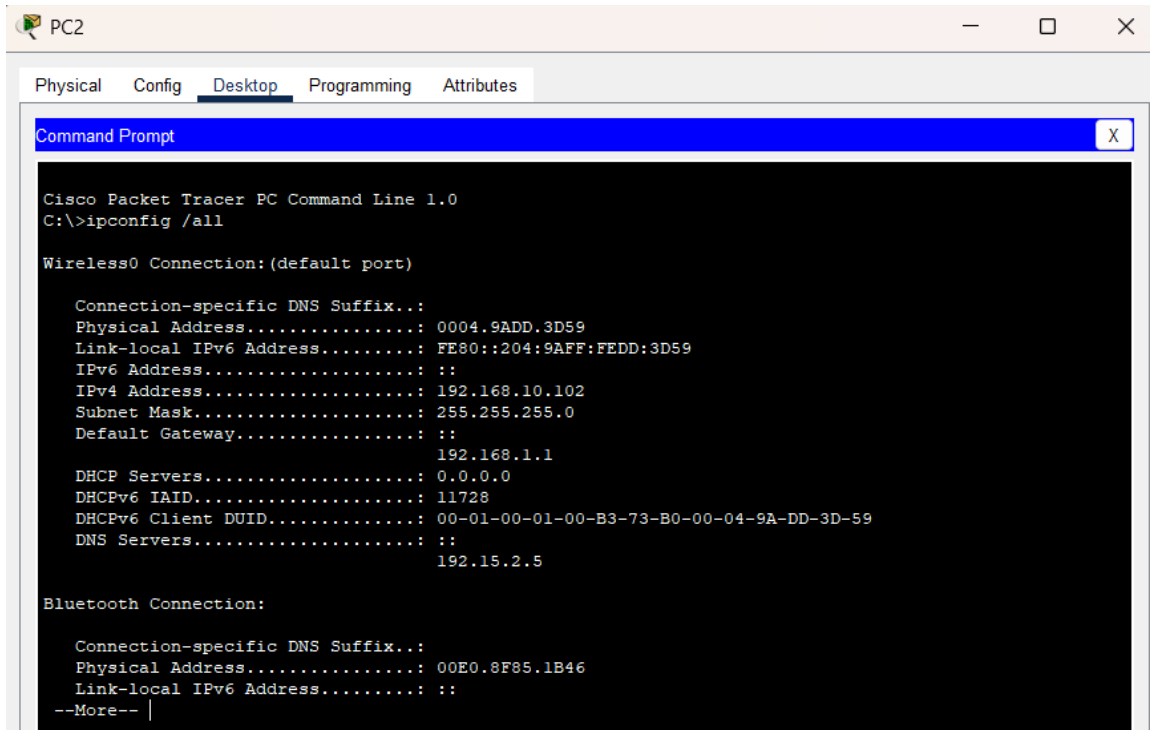
Wireless0 Connection:(default port)

Connection-specific DNS Suffix...:
Physical Address.....: 0090.210A.70A5
Link-local IPv6 Address.....: FE80::290:21FF:FE0A:70A5
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.101
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        192.168.1.1
DHCP Servers.....: 192.168.1.1
DHCPv6 IAID.....: 11940
DHCPv6 Client DUID.....: 00-01-00-01-D2-06-CD-3B-00-90-21-0A-70-A5
DNS Servers.....: ::
                        192.15.2.5

Bluetooth Connection:

Connection-specific DNS Suffix...:
Physical Address.....: 0001.97D4.61E3
Link-local IPv6 Address.....: ::
--More-- |
```

PC1 ipconfig



PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

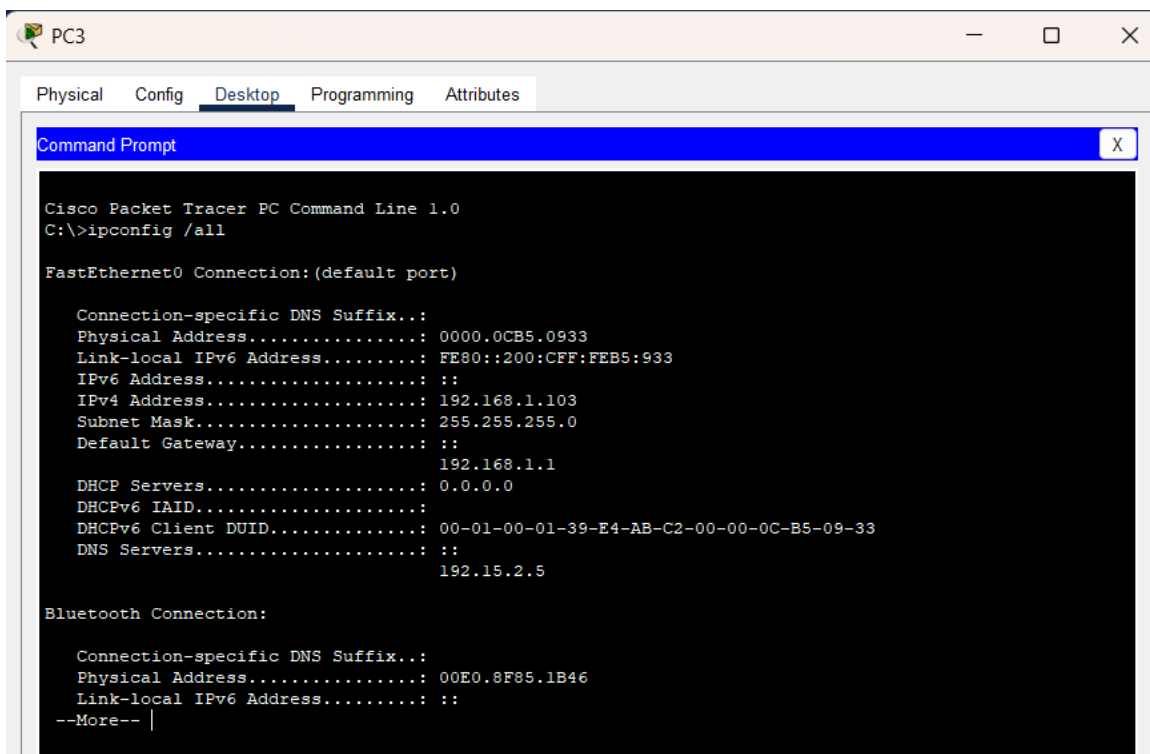
Wireless0 Connection: (default port)

Connection-specific DNS Suffix...:
Physical Address.....: 0004.9ADD.3D59
Link-local IPv6 Address.....: FE80::204:9AFF:FEDD:3D59
IPv6 Address.....: ::
IPv4 Address.....: 192.168.10.102
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        192.168.1.1
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....: 11728
DHCPv6 Client DUID.....: 00-01-00-01-00-B3-73-B0-00-04-9A-DD-3D-59
DNS Servers.....: ::
                        192.15.2.5

Bluetooth Connection:

Connection-specific DNS Suffix...:
Physical Address.....: 00E0.8F85.1B46
Link-local IPv6 Address.....: ::
--More-- |
```

PC2 ipconfig



PC3

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

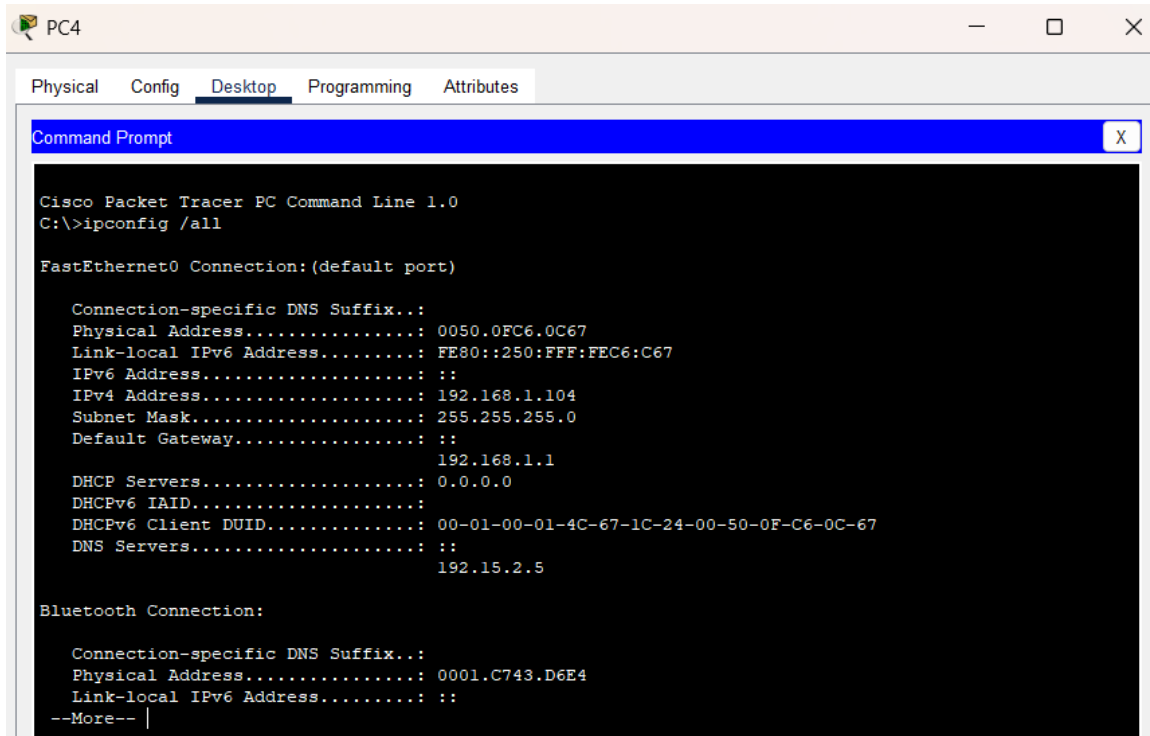
FastEthernet0 Connection: (default port)

Connection-specific DNS Suffix...:
Physical Address.....: 0000.0CB5.0933
Link-local IPv6 Address.....: FE80::200:CFF:FEB5:933
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.103
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        192.168.1.1
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-39-E4-AB-C2-00-00-0C-B5-09-33
DNS Servers.....: ::
                        192.15.2.5

Bluetooth Connection:

Connection-specific DNS Suffix...:
Physical Address.....: 00E0.8F85.1B46
Link-local IPv6 Address.....: ::
--More-- |
```

PC3 ipconfig



The screenshot shows a PC window titled 'PC4' with tabs for Physical, Config, Desktop, Programming, and Attributes. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the output of the 'ipconfig /all' command, detailing network configuration for FastEthernet0 and Bluetooth connections.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...: 
    Physical Address.....: 0050.0FC6.0C67
    Link-local IPv6 Address.....: FE80::250:FFF:FEC6:C67
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.1.104
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                        192.168.1.1
    DHCP Servers.....: 0.0.0.0
    DHCPv6 IAID.....: 
    DHCPv6 Client DUID.....: 00-01-00-01-4C-67-1C-24-00-50-0F-C6-0C-67
    DNS Servers.....: ::
                        192.15.2.5

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Physical Address.....: 0001.C743.D6E4
    Link-local IPv6 Address.....: ::
    --More-- |
```

PC4 ipconfig

Analysis of IP Information

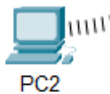
Correct configuration should follow the pattern:

- IP Address: 192.168.1.x
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

Any deviation identifies the misconfigured PC.

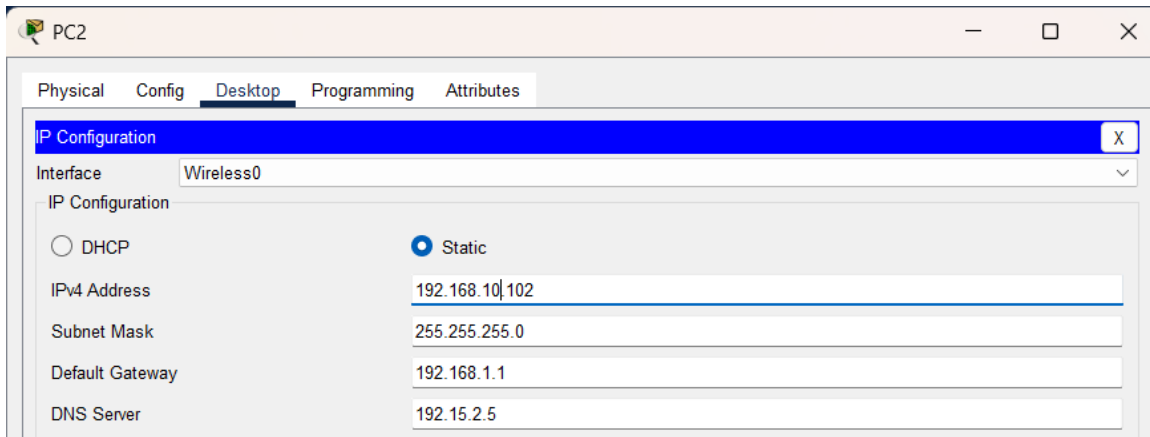
4. Part 2: Correct Any Misconfigurations

a. Select the PC that has incorrect IP information.

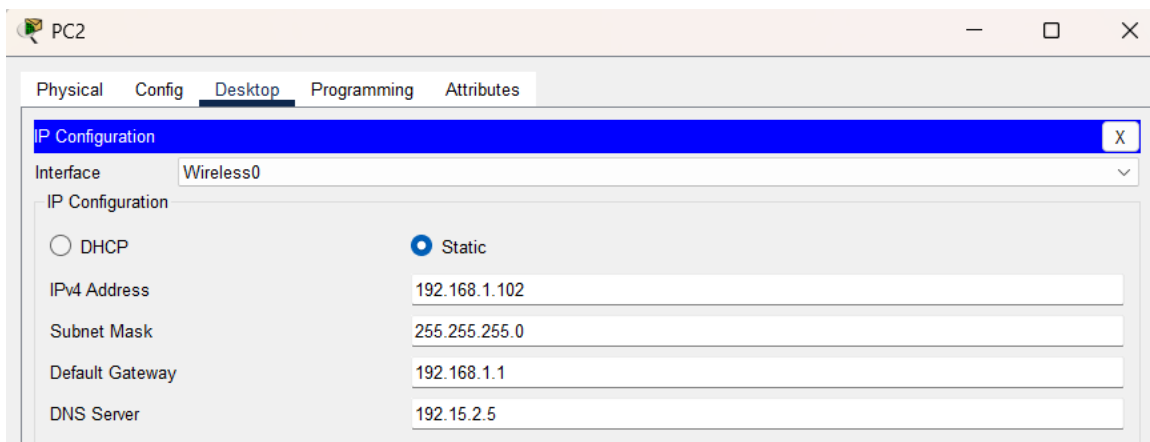


b. Open Desktop > IP Configuration.

c. Enter correct values for IP address, subnet mask, and default gateway.



Before correct ip settings



Corrected IP Settings

5. Observations & Conclusion

After correcting the addressing issue, the PC should successfully reach www.cisco.pka. The `ipconfig` command is a primary troubleshooting tool for identifying IP configuration issues.